

HATEM ISMAIL SHALABY

ML & WORKFLOW AUTOMATION ENGINEER · CONTACT CENTER INTELLIGENCE

DevOps / SRE / Platform Engineering • Real-Time Adherence • Predictive Staffing

AI Obour City, Egypt | +20 100 337 2510 | hatemismail2011@gmail.com | [LinkedIn](#) | github.com/TomyShelby79

ENGINEERING PROFILE

28 years in operations. The last 14 building AI-driven automation for contact centers at ByteDance, Vodafone, and Uber.

I engineer production AI systems that replace operational chaos with self-sustaining automation. My work bridges frontline reality — aviation crisis management, retail logistics — with technical execution: NLP pipelines, Erlang C forecasting. I don't just write code; I build systems that run without me, delivering measurable ROI in live environments.

CORE COMPETENCIES & PLATFORMS

- **WFM & RTA Domains:** Real-Time Adherence, Predictive Staffing, Erlang C Forecasting, Churn Prediction, Anomaly Detection, SOP Automation.
- **Platforms & Tools:** Replaced NICE IEX, Genesys, Calabrio with custom builds. Integrated Zendesk, Salesforce, Jira, Confluence, Notion, Airtable, Zapier/Make.
- **Technical Stack:** Python (Pandas, NumPy, scikit-learn, FastAPI, Streamlit), SQL (Window Functions, CTEs), Excel VBA, Git/GitHub, Docker, Kubernetes, GCP, Azure, CI/CD (GitHub Actions).
- **Architecture:** Clean Architecture, Microservices, API Development, Containerization, Multi-cloud Infrastructure.

PROFESSIONAL EXPERIENCE

AI Workflow Automation Engineer | ByteDance (TikTok LIVE)

2025 – Present

- **Throughput:** Engineered AI-assisted workflow tools scaling pilot output 186% (14 → 40 tasks/day) with zero headcount increase.
- **Efficiency:** Deployed automated macro libraries reducing Average Handle Time (AHT) and standardizing agent performance across high-volume queues.
- **Proactivity:** Built SQL anomaly-detection pipelines to preempt KPI degradation, shifting operations from reactive response to structured early intervention.

Transition note: converted from BPO vendor (TTEC) to direct engineering role.

Process Automation & Systems Analyst | TTEC

2023 – 2025

- **Optimization:** Redesigned end-to-end SOPs via root-cause auditing, cutting AHT by 18% and lifting First-Contact Resolution (FCR) by 22%.
- **Quality:** Maintained 95%+ CSAT and 92% AQA through systematic process enforcement and embedded feedback loops in operational tooling.

WFM Engineering & Analytics Lead | Vodafone UK Limited

2020 – 2023

- **Forecasting:** Replaced legacy WFM tools (NICE/Genesys) with custom Erlang C forecasting engines, predicting FTE needs within 8% variance and driving schedule adherence to 92%.
- **Automation:** Automated KPI dashboards (VBA/Sheets), recovering 12 FTE-hours weekly and tracking 70+ weekly cases.
- **Strategy:** Translated model outputs into operational strategy, increasing FCR by 20% and CSAT by 8 points across global cohorts.

Data & Operations Automation Analyst | Uber

2015 – 2020

- **Reporting:** Engineered real-time SQL pipelines for KPI tracking, eliminating 8 hours of manual weekly reporting.

- **Retention:** Developed cohort analysis models identifying high-churn segments; interventions improved activation rates by 25%.
- **Scaling:** Authored 6 core SOPs enabling agent team scaling from 15 to 60+ without process regression.

Technical Support Representative | Vodafone UK Limited

2012 – 2015

- **Frontline Execution:** Managed 50+ daily calls with full accountability for AHT, CSAT, FCR, and Adherence SLAs.
- **Insight:** Identified systemic operational patterns at the frontline, surfacing critical issues to leadership for process correction.

CAREER FOUNDATION (1998 – 2012)

Operational leadership across aviation, retail, and education IT — crisis-tested in the 2011 regional transition.

- **Senior Operations Officer** | Air Berlin, Hurghada | 2009 – 2011 — Directed ground ops during the 2011 revolution; maintained SLA compliance under extreme pressure.
- **Ground Handling & Traffic Officer** | Hurghada Int'l Airport | 2007 – 2009 — Executed real-time traffic management with strict aviation safety SLAs.
- **IT Help Desk & ICT Coordinator** | Nefertari American School | 2005 – 2007 — Managed hybrid IT operations ensuring 99.9% uptime.
- **Director, International Affairs** | Shorouk International Bookshop | 2003 – 2005 — Led P&L management and vendor relations for international logistics.
- **International Affairs Coordinator** | Shorouk International Bookshop | 1998 – 2003 — Coordinated cross-border business operations and supply chain.

PRODUCTION PORTFOLIO: THE HELIX ECOSYSTEM

Four interconnected, production-grade automation systems. Not demos.

SYSTEM	FUNCTION	IMPACT
Helix_OPS	Intake-to-readiness pipeline converting raw requirements into Erlang C schedules and Notion SOPs.	< 60 min setup time for new campaigns.
Helix_Prime	Clean Architecture FastAPI system for real-time workforce optimization.	Zero manual intervention post-deployment; auto-adjusts forecasts.
Helix_Sonar	NLP-driven churn prediction and sentiment monitoring.	48–72 hr early warning for at-risk accounts.
Helix_Watch	Real-time adherence dashboard with automated anomaly detection.	Flags degradation before KPI reports register.

GitHub: github.com/TomyShelby79 (Helix_OPS, Helix_Prime, Helix_Sonar, Helix_Watch)

EDUCATION

- **Bachelor’s in Managerial Sciences (Computer Section)** — Sadat Academy for Management Sciences
- **Business Analytics Nanodegree** — Udacity